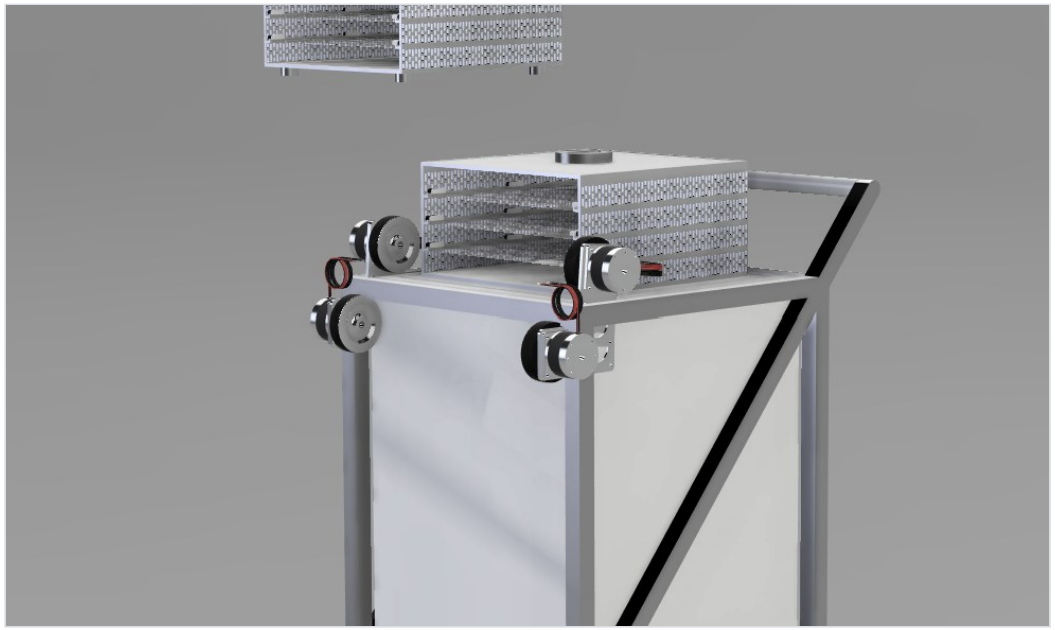
ENGINEERING PORTFOLIO · 2026

Justin Kang

I test, integrate, and design physical systems — from robot pick-and-place cells to a retrofitted excavator — and I care most about finding out *why* something fails.



SKYLINE TRANSPORT · OVERHEAD GANTRY HANDING A STORAGE UNIT TO ITS CART (TEAM CAD)

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02 **EXCA-Scan**
RC excavator retrofitted to scan objects into to-scale CAD · hardware control + learned policy

04 **Skyline Transport**
Cleanroom overhead gantry + cart · decision matrices, iteration, stress check, BOM

06 **Test & Validation Work**
Robotics QA at Ambi Robotics · policy benchmarking at Berkeley AI Research

07 **Software Builds**
BuzzBuddy · UrbanPilot · Flux

Each project is laid out the same way: the context and constraints, what my role was, then the key decisions — what I did, how, and why I chose it over the alternatives.

EXCA-Scan

act-athon physical AI hackathon
Sept 25–27, 2026 · team build

CONTEXT

At a physical-AI hackathon we turned a toy RC excavator into an object scanner: the bucket comes off, a camera goes on, and the machine drives itself around an object (e.g. a soda can) to capture every angle — producing a to-scale CAD model you could 3D-print or design features onto.

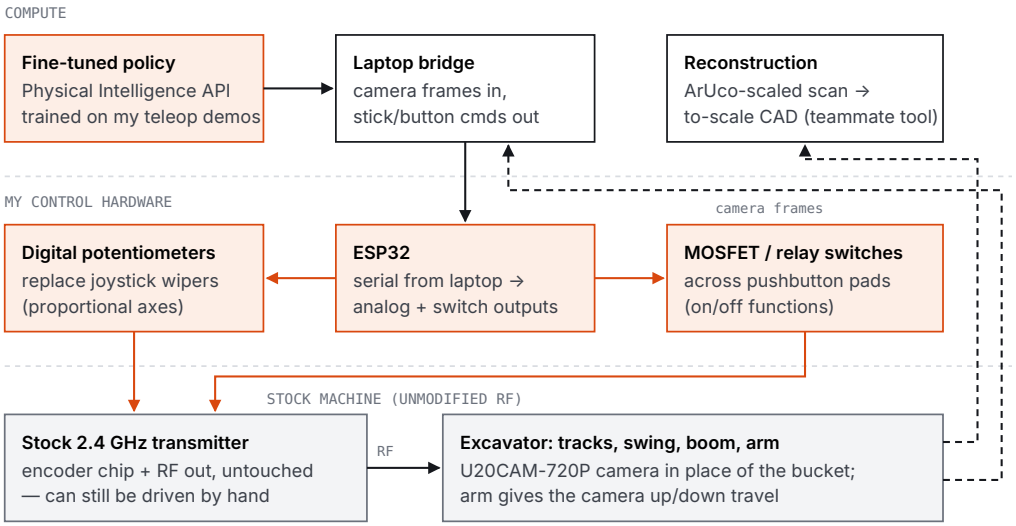
CONSTRAINTS

- Stock 2.4 GHz toy transmitter — joystick potentiometers and pushbuttons into an encoder chip, **no digital or serial interface**.
- Access to Physical Intelligence's model suite (incl. π0.7) through their API — but no on-device GPU (my laptop is Apple Silicon, no CUDA).
- Hackathon clock: whatever we built had to be debuggable on a breadboard in a weekend.

MY ROLE

I owned **control and autonomy**: getting a computer to drive the excavator through its own transmitter, collecting teleoperated demonstrations, and fine-tuning the policy that moves the camera around the object. Teammates handled the bucket-to-camera mount and the reconstruction tool.

SYSTEM ARCHITECTURE



ORANGE = WHAT I BUILT. DASHED = CAMERA FEEDBACK TO THE POLICY AND TO RECONSTRUCTION.

Key decisions — what, how, why

control · data · calibration

D.01 Drive the transmitter in hardware instead of modifying the RF link or receiver

How: an ESP32 drives digital potentiometers wired in place of the joystick wipers, and MOSFET/relay switches across the button contacts — so the transmitter's own encoder chip thinks a person is holding the sticks.

WHY → The toy board had no digital interface, and reverse-engineering its 2.4 GHz protocol was not a weekend-sized problem. Injecting at the stick level reuses the stock radio unchanged, keeps a human-drivable fallback, and turns every control channel into something I could test with a multimeter.

D.02 Learn the scanning motion from teleoperated demonstrations

How: I drove the scanning pattern myself through the same control path to record demonstrations, then fine-tuned a Physical Intelligence model on them through its API.

WHY → Toy-grade actuators drift and stick, so a hand-scripted orbit breaks as soon as the object or surface changes. Demos recorded through the exact hardware the policy would use mean the model learns the machine's real behavior — and the API kept training off a laptop with no GPU.

D.03 Fix scale with ArUco markers, and swap our pipeline for a teammate's tool

How: printed ArUco markers placed around the object give known reference dimensions; we dropped a COLMAP / OpenMVS / Open3D chain for a reconstruction tool a teammate had already built.

WHY → A photo-only reconstruction has no absolute scale — useless if the goal is a part you can print. Markers make the CAD dimensionally true. Switching tools cut integration risk late in the event.

OUTCOME

- End-to-end loop working: **policy → ESP32 → stock transmitter → excavator**, with the camera feeding both the policy and reconstruction.
- Produced **to-scale CAD models** of scanned objects.
- Honest gap: none were 3D-printed before the deadline, so the "scan → physical part" claim is untested.

WHAT I'D DO NEXT

- **Measure scan accuracy** — print a scanned calibration block and check dimensions with calipers against the original.
- **Characterize the control path** — log commanded vs. actual joint motion per channel to find dead-band and lag, then compensate.
- **Move from breadboard to a soldered board** with strain relief; loose jumpers were our most common failure while debugging.
- Add a hardware kill switch independent of the ESP32.

SKILLS SHOWN

Embedded control (ESP32) · analog signal injection · hardware debugging · imitation learning / policy fine-tuning · fiducial-based calibration · system integration under time pressure

Skyline Transport

ASME UC Berkeley CADathon · Apr 2025
2nd place (tie) of 12+ teams · 12-hour event · team of 3

CONTEXT

Prompt: improve material handling in a semiconductor fab. Our answer is an overhead gantry that carries a storage unit between bays, plus a separate motorized cart that hands the unit up to it.

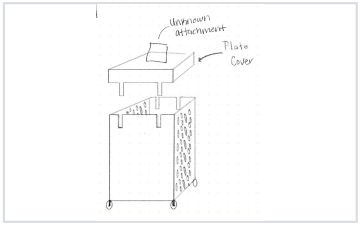
CONSTRAINTS

- **ISO 14644 cleanroom** — smooth, sealed surfaces; no particle traps.
- **Vibration and ESD** can scrap wafers in transit.
- Must reconfigure to different fab layouts.
- Buildable from catalog parts — final BOM **$2,309.85**.

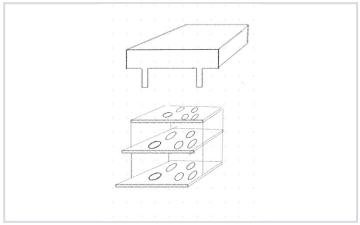
MY ROLE

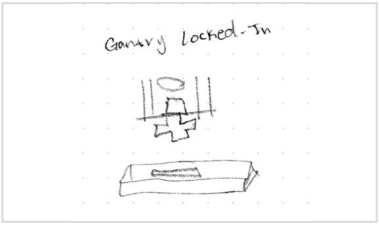
I led **research, design ideation, the decision matrices, and the bill of materials**, and ran the static stress check on the tray. Jacob Reyes led CAD; Yahany Francisco worked on design process and brainstorming.

FROM SKETCH TO SELECTION



Early cart + unknown attachment


Pull-out tray concept



"Gantry locked-in" lock sketch

Matrix 1 — overall system

OPTION	USAB. ×1.70	EFFIC. ×1.90	COST ×1.00	MNVR. ×1.87	SAFETY ×1.99	SCORE
Ceiling	**7**	**7**	**4**	**6**	**8**	**71.37**
Floor	9	7	9	0	8	**67.02**

Conveyor options. Also scored: design-change feasibility ×1.60, cleanliness ×1.80.

D.01 · GO OVERHEAD, THEN ADD A CART

The floor conveyor was cheaper and easier to use but scored **0 on maneuverability** — it can't follow a fab's changing layout and it takes up controlled floor space.

WHY → Safety, efficiency and maneuverability were weighted ~2× cost, because a damaged wafer lot costs far more than the hardware. The cart adds what a ceiling rail can't: moving material in and out of the room.

Matrix 2 — gantry-to-unit attachment

OPTION	EFFIC.	COST	CLEAN.	SAFETY	SCORE
Mech. lock	**9**	**7**	**3**	**9**	**79.79**
Magnetic	7	3	6	8	**76.48**
Air suction	4	4	5	8	**60.70**

Same seven weighted criteria; four columns shown.

D.02 · MECHANICAL LOCK OVER MAGNETIC OR SUCTION

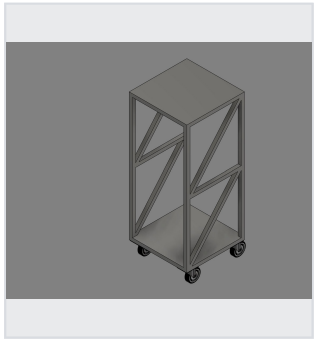
It scored highest on efficiency and safety, but **worst on cleanliness (3/10)** — a real cost in an ISO cleanroom.

WHY → We took the trade and designed around it: sealed extrusions, slot covers, and covered fasteners so the lock doesn't become a particle trap.

Iteration, verification, and cost

CAD renders are team work (CAD led by J. Reyes)

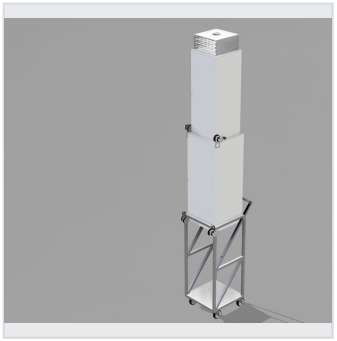
D.03 · MOVE THE LIFT OUT OF THE GANTRY HEAD



Cart V1 — storage fixed to a push cart



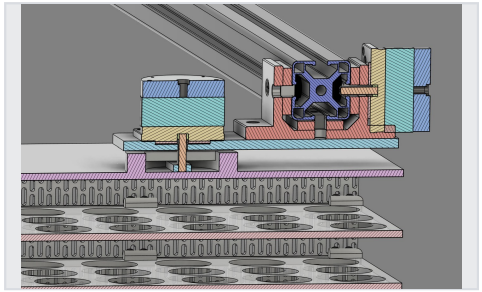
Cart V2 — separate storage unit



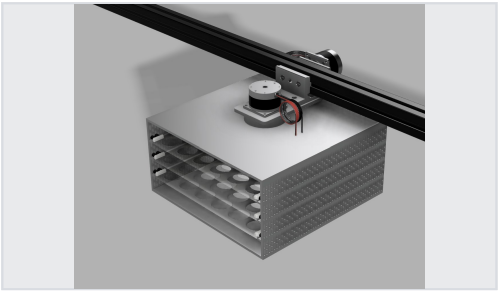
Final — motorized elevator raises the unit

We first planned to hoist the storage unit from the gantry head. WHY WE CHANGED → every motor and mechanism on the head is mass the rail has to carry and another point of failure overhead. Putting the lift in the cart lowered the overhead load and kept heavy motion near the floor, where it is safer to service.

GANTRY HEAD + LOCKING MECHANISM

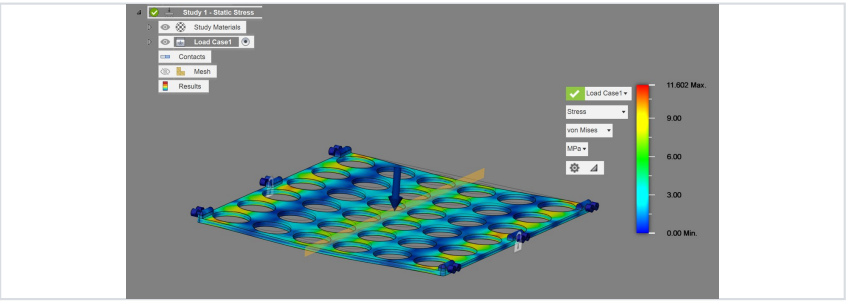


Cross-section: stepper rotates a key into the unit's T-slot mount



Head on 6061-T6 T-slot rail

VERIFICATION · STATIC STRESS ON THE TRAY



Static stress study of the perforated tray under load

The waffle-perforated tray saves mass but removes material where load concentrates, so I checked it before committing: **5–50 lb payloads with a factor of safety above 4**.

BILL OF MATERIALS — $2,309.85

ITEM	QTY	TOTAL	SHARE
Stepper motor	**8**	**$924.16**	**40%**
T-slot roller	30	$362.10	16%
Aluminum sheet	1	$313.96	14%
T-slot slider	3	$259.35	11%
T-slot rail	3	$244.68	11%
Aluminum rect. tube	3	$172.20	7%
Acrylic sheet	1	$33.40	1%

All parts sourced from McMaster-Carr.

WHAT I'D CHANGE

Steppers are 40% of cost. Next revision: audit which of the 8 motorized axes truly need a stepper, and test the tray physically against the FEA instead of trusting the simulation alone.

Testing real robots

Proprietary systems — described at a high level, no internal data or images

AMBI ROBOTICS · ROBOTICS QA INTERN · FALL 2026–PRESENT

AmbiSort & AmbiStack pick-and-place systems

On-site in Berkeley three days a week, working with the CTO and software team.



- **Defect triage, 1–2 tickets a day** in Jira and Asana, working alongside an in-house AI bug-tracking agent.
- **Data-driven diagnosis** — reading system logs and the team's Grafana dashboards to separate one-off faults from recurring ones.
- **Found a recurring gantry collision fault**, traced it to a safety-system configuration issue, reported it with suggested fixes; the lead software engineer took the fix forward.
- **Led 3 customer demos**, explaining the conveyor system and operator HMI to non-technical users.

WHY IT MATTERS → A single collision reads as noise; the same collision across days is a design or configuration problem. Trending faults in the dashboards before filing is what made this one actionable.

BERKELEY AI RESEARCH · RAIL LAB (PROF. SERGEY LEVINE) · OCT 2024–MAY 2026

Benchmarking learned manipulation policies

Undergraduate research assistant, ~20 months.

2 POLICIES A/B TESTED

10 BENCHMARK TASKS

~5 PHYSICAL EPISODES / DAY

- Ran **structured manipulation benchmarks on a Franka Emika arm** to measure how robust ML policies are to changes in the task setup.
- Built **Python / Pandas evaluation pipelines** to turn episode logs into comparable per-task results.
- **Cleaned, labeled, and validated** large-scale robot training datasets.
- Documented A/B results for the research team.

WHY IT MATTERS → At roughly five physical episodes a day, every trial is expensive. That forces a disciplined test plan — the same setup and scoring every time — so a difference in results reflects the policy, not how the test was run.

THROUGH-LINE

Both roles are the same job at different scales: define what "working" means, measure it consistently on real hardware, and turn failures into evidence an engineer can act on.

Software builds

Hackathon builds, 2026
code + writeups at justin-kang.vercel.app

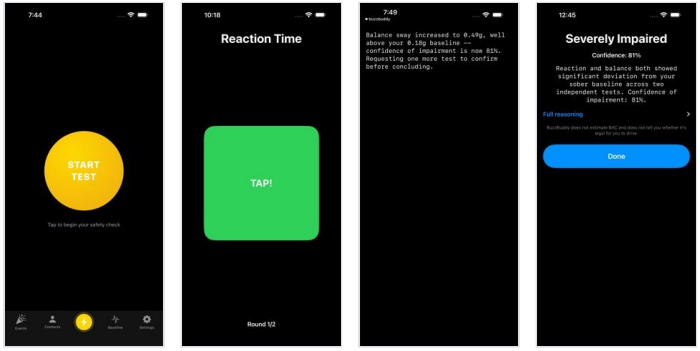
BuzzBuddy · Jul 2026

3RD PLACE · AI FOR SOCIAL GOOD · MLH × DIGITALOCEAN

iOS app that flags impairment by comparing reaction, balance, and memory tests against **your own sober baseline** instead of a population BAC estimate.

My part: the FastAPI / SQLAlchemy backend and the agentic tool-calling loop (retrieve baseline → request test → analyze deviation → update confidence), wired into the team's SwiftUI app.

WHY → A tool loop lets the examiner ask for one more test only when evidence is ambiguous, and every verdict cites a specific percent deviation — not an opaque score.



Check-in → reaction test → examiner reasoning → verdict (never "safe to drive")

UrbanPilot · Jun 2026 · UC Berkeley AI Hackathon

Enter any real address and a planning goal; four specialist Claude agents (climate, housing, accessibility, urban design) analyze it in parallel against public data, and Midjourney renders 2040 / 2075 streetscapes from the real Street View.

My part: primary author of the React/Leaflet frontend, Node/Express backend, and the multi-agent coordinator. A teammate built the verified-data integrations.

Measured, not estimated: my prompt-compression layer cut input tokens **27.8% / 35.4% / 33.7%** (housing / climate / accessibility agents) on real API calls, with every response still valid JSON citing the source figures.

Flux · Apr 2026 · LA Hacks · team of 4, 36 hrs

Voice-first automation on Meta Ray-Ban glasses: one spoken command becomes a multi-step workflow across 12+ services (Gmail, Calendar, Slack, Notion, Fetch.ai).

WHY → Spoken commands are messy and incomplete, so the Gemini parser emits workflow JSON against a **closed-world action schema**, and a validate → repair loop fixes missing or invalid fields before anything executes. Predictable output beat a free-form agent for reliability.

WHY THESE ARE HERE

Test and integration work is half software: logging, data pipelines, and tooling that turns a pile of runs into a clear answer. These builds are where I practice that.